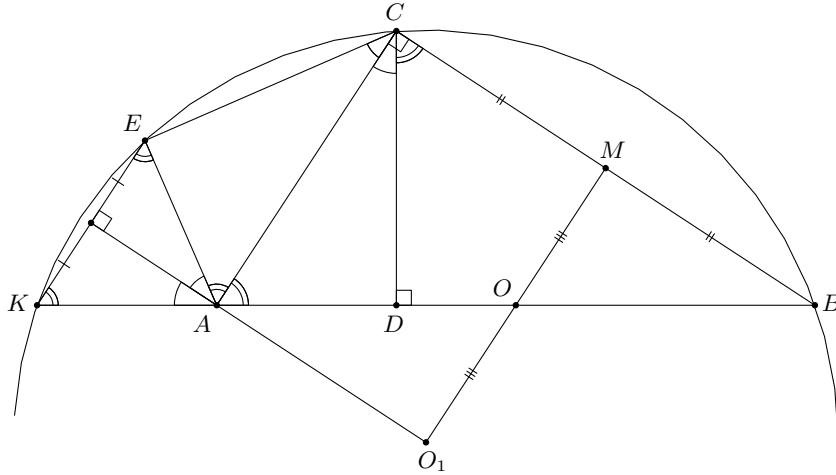


Problem 45. Let ABC be a right-angled triangle with $\angle ACB = 90^\circ$. Denote by

- D the foot of the perpendicular from C to AB ;
- E and F the reflections of D about AC and BC , respectively;
- O_1 and O_2 the circumcentres of $\triangle ECB$ and $\triangle FCA$, respectively.

Prove that $AB = 2 \cdot O_1O_2$.

Solution.

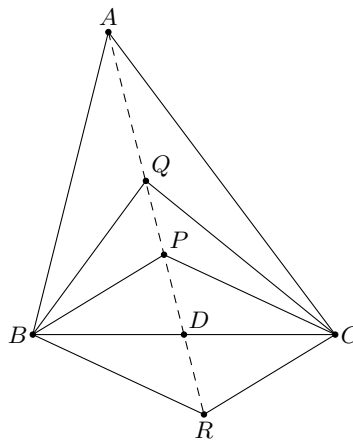


Let $\angle ABC = \alpha$ and $\angle BAC = 90^\circ - \alpha = \beta$. Then $\angle BCD = \angle EAC = \beta$ and $\angle ACD = \angle ACE = \alpha$. Let K be the second intersection point of the circumcircle of $\triangle ECB$ and the line AB . Then $\angle AKE = \angle KEA = \beta$, and so $\triangle AKE$ is an isosceles triangle.

Now observe that O_1 is the intersection point of the perpendicular bisectors of EK and BC . Let O and M be the midpoints of AB and BC . Then M, O and O_1 are collinear. By angle chasing, we see that AO_1MC is a rectangle. Moreover, since $\triangle OMB \cong \triangle OO_1A$, we have $OM = OO_1$. Hence $AC \parallel OO_1$ and $AC = 2 \cdot OO_1$. Similarly, we have $BC \parallel OO_2$ and $BC = 2 \cdot OO_2$. Therefore $\triangle ABC \sim \triangle O_1O_2O$ and so $AB = 2 \cdot O_1O_2$. $\square$

Problem 46. In an acute-angled triangle ABC , let D be the midpoint of BC and let P be a point on the segment AD . The interior angle bisectors of $\angle ABP$ and $\angle ACP$ intersect at Q . If $BQ \perp QC$, prove that Q lies on the segment AP .

Solution.



By angle chasing, we have

$$\begin{aligned} \angle BAC + \angle BPC &= 2\angle BAC + \angle ABP + \angle ACP \\ &= 2(\angle BAC + \angle ABQ + \angle ACQ) \\ &= 2\angle BQC \\ &= 180^\circ. \end{aligned}$$

Let R be a point on PD such that $PD = DR$. Since $BD = DC$ and $PD = DR$, we conclude that $BPCR$ is a parallelogram. Therefore $\angle BRC = \angle BPC = 180^\circ - \angle BAC$ which means that $BACR$ is a cyclic quadrilateral. Since $BD = DC$, the areas of the triangles ABR and ACR are equal. We also have $\angle ABR + \angle ACR = 180^\circ$ and so we get $AB \cdot BR = AC \cdot CR$ which shows that

$$\frac{AB}{AC} = \frac{CR}{BR} = \frac{BP}{CP}.$$

Let $Q' = BQ \cap AP$. Observe that since $\triangle ABC$ is acute, Q' must be on the line segment AP . Now, by the Angle Bisector Theorem, we get

$$\frac{AC}{CP} = \frac{AB}{BP} = \frac{AQ'}{PQ'}$$

and hence CQ' should also bisect of the $\angle ACP$, i.e. C, Q, Q' are collinear. This is possible only if $Q = Q' \in AP$. $\square$

Problem 47. Suppose 100 points in the plane are coloured using two colours, red and white, such that each red point is the centre of a circle passing through at least three white points. What is the least possible number of white points?

Solution. Let n be the number of white points. Then we can draw at most $\binom{n}{3}$ circles. Thus the number of white and red points together is at most $n + \binom{n}{3}$. We observe that $9 + \binom{9}{3} = 93$ and $10 + \binom{10}{3} = 130$. Thus $n \geq 10$. We show that $n = 10$ works. Take any 10 points such that no three are collinear and the circumcentres of the triangles formed by any 3 of these points are pairwise distinct and different from these 10 points. (We can choose the points one by one. Each time we only need to avoid a finite number of points, lines and circles, so there must be an appropriate choice.) Then we get $\binom{10}{3} = 120$ points as centre of distinct circles. Choose any 90 points from them and colour them red and colour the original 10 points white. We get 100 points of which 10 are white and remaining 90 are red. Each of these 90 red points is the centre of a circle passing through some three white points. $\square$

Problem 48. Find all ordered pairs (m, n) of integers so that the following quartic equation in x given by

$$x^4 + 2mx^2 + nx + m^2 - 36 = 0 \quad (*)$$

has four integer roots. The roots are counted with multiplicity, i.e. double roots are counted twice, triple roots are counted thrice, etc.

Solution. Let $a, b, c, -a - b - c$ be the four integer roots of $(*)$. Then

$$2m = ab + bc + ca - a(a + b + c) - b(a + b + c) - c(a + b + c) = -(a^2 + b^2 + c^2 + ab + bc + ca)$$

and

$$-n = abc - ab(a + b + c) - bc(a + b + c) - ca(a + b + c).$$

Hence

$$-4m = (a + b)^2 + (b + c)^2 + (c + a)^2 \quad (**)$$

and

$$n = (a + b)(b + c)(c + a).$$

From $(**)$, we see that $a + b, b + c$ and $c + a$ are even. Let $a + b = 2u, b + c = 2v$ and $c + a = 2w$ for some integers u, v and w . Then

$$\begin{cases} -m = u^2 + v^2 + w^2, \\ n = 8uvw, \end{cases} \quad \text{and} \quad \begin{cases} a = u - v + w, \\ b = u + v - w, \\ c = -u + v + w. \end{cases}$$

Since $m^2 - 36 = -abc(a + b + c)$, it follows that

$$(u^2 + v^2 + w^2)^2 - 36 = -(u - v + w)(u + v - w)(-u + v + w)(u + v + w) = u^4 + v^4 + w^4 - 2u^2v^2 - 2v^2w^2 - 2w^2u^2$$

and so

$$u^2v^2 + v^2w^2 + w^2u^2 = 9.$$

Now we consider only the following cases (because their permutations yield the same answer):

- When $(u^2v^2, v^2w^2, w^2u^2) = (9, 0, 0)$, we have $uv = \pm 3$ and $w = 0$, and so $m = -10$ and $n = 0$. Then $(*)$ becomes $x^4 - 20x^2 + 64 = 0$. Clearly, this has integer roots because $x^4 - 20x^2 + 64 = (x - 4)(x + 4)(x - 2)(x + 2)$ and so satisfies the problem condition.
- When $(u^2v^2, v^2w^2, w^2u^2) = (4, 4, 1)$, we have $u^2 = w^2 = 1$ and $v^2 = 4$, and so $m = -6$ and $n = \pm 16$. Then $(*)$ becomes either $x^4 - 12x^2 + 16x = 0$, which has integer roots because $x^4 - 12x^2 + 16x = x(x - 2)^2(x + 4)$, or it becomes $x^4 - 12x^2 - 16x = 0$, which also has integer roots because $x^4 - 12x^2 + 16x = x(x + 2)^2(x - 4)$. So both satisfy the problem condition.

Therefore the required ordered pairs (m, n) are $(-10, 0), (-6, 16), (-6, -16)$. $\square$